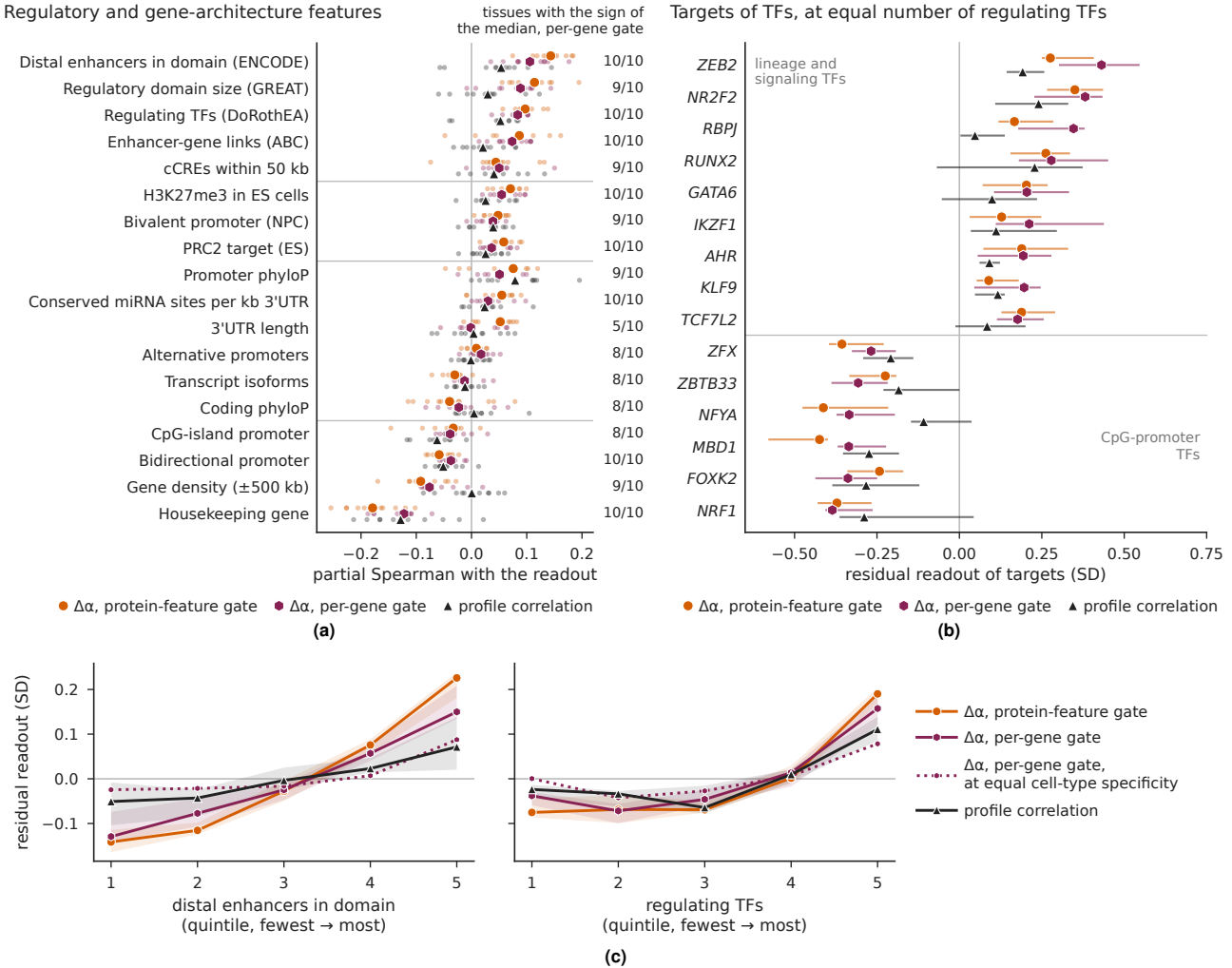

Regulatory and gene-architecture features
tissues with the sign of the median, per-gene gate
Distal enhancers in domain (ENCODE)
Regulatory domain size (GREAT)
Regulating TFs (DoRothEA)
Enhancer-gene links (ABC)
cCREs within 50 kb
H3K27me3 in ES cells
Bivalent promoter (NPC)
PRC2 target (ES)
Promoter phyloP
Conserved miRNA sites per kb 3'UTR
3'UTR length
Alternative promoters
Transcript isoforms
Coding phyloP
CpG-island promoter
Bidirectional promoter
Gene density (±500 kb)
Housekeeping gene
10/10
9/10
10/10
10/10
9/10
10/10
9/10
10/10
9/10
10/10
5/10
8/10
8/10
8/10
8/10
10/10
9/10
10/10
−0.2 −0.1 0.0 0.1 0.2
partial Spearman with the readout
Δα, protein-feature gate
Δα, per-gene gate
profile correlation
(a)
Targets of TFs, at equal number of regulating TFs
lineage and signaling TFs
ZEB2
NR2F2
RBPJ
RUNX2
GATA6
IKZF1
AHR
KLF9
TCF7L2
ZFX
ZBTB33
NFYA
MBD1
FOXK2
NRF1
CpG-promoter TFs
−0.50 −0.25 0.00 0.25 0.50 0.75
residual readout of targets (SD)
Δα, protein-feature gate
Δα, per-gene gate
profile correlation
(b)
residual readout (SD)
0.2
0.1
0.0
−0.1
1 2 3 4 5
distal enhancers in domain (quintile, fewest → most)
1 2 3 4 5
regulating TFs (quintile, fewest → most)
Δα, protein-feature gate
Δα, per-gene gate
Δα, per-gene gate, at equal cell-type specificity
profile correlation
(c)